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Joint Monitoring of Acoustic Emission and Fracture Geometry in Glutenite Hydraulic Fracturing Based on High-Frequency and Low-Frequency Das Data

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Hydraulic fractures are typically caused by the combined effects of external and in-situ stresses and manifest as a complex fracture networks of natural and artificial fractures (Chen et al., 2021; Schultz et al., 2020). However, these fractures are difficult to detect and observe in reservoirs for lack of full complete three-dimensional velocity model and monitoring system. Acoustic emission events (AEs) are defined as transient elastic waves that characterize the nucleation and evolution of rock fractures during laboratory-scale hydraulic fracturing (Li et al., 2023). Piezoelectric transducers (PZTs) are typically used to detect this motion and convert it into an electrical signal, which can then be used for AE localization. However, since the voltage cannot reflect the stress state of the fractures, using PZTs alone cannot simultaneously monitor AE activity and characterize crack features.

Distributed Acoustic Sensing (DAS) is a revolutionary seismic monitoring technology that converts fiber-optic cables into dense sensors to detect and monitor a wide range of vibrations (Daley et al., 2013; Lindsey and Martin, 2021), including acoustic waves, seismic waves and even ambient noise. The operation of DAS is relatively simple: the interrogator unit (IU) emits laser pluses and receive backscattered Rayleigh light from the cables for measurement. In addition, DAS requires no electronic devices in the sensing cables and is resistant to high temperatures and corrosion. As a result, it has been widely applied in various fields, including oil and gas development, geothermal reservoir monitor, civil engineering, and smart city infrastructure.

While Distributed Acoustic Sensing (DAS) has proven to be a powerful tool for independently monitoring acoustic emission (AE) events and characterizing fracture behavior during hydraulic fracturing, integrated analyses remain relatively limited. Hull et al. (2017) demonstrated a spatial and temporal correlation between microseismicity and DAS-recorded strain along the fiber, suggesting that the strain deformation boundary effectively constrains the extent of induced seismicity. Similarly, Starr and Jacobi (2017) reported a significant association between seismic locations and relative strain, with most microseismic

events triggering measurable strain perturbations. Li et al. (2020) identified multiple new tensile-mode fractures from low-frequency DAS strain signals, and subsequently converted these strain responses into microseismic signals. Their results revealed a strong spatial correlation between strain-inferred fractures and microseismic locations, providing compelling evidence of strain-induced fracture formation. However, these studies primarily focus on field-scale hydraulic fracturing in reservoirs and lack high-resolution validation at the scale of moderate- to small-magnitude seismicity. This limitation hinders accurate assessment of fracture-induced seismic criticality. Applying DAS for simultaneous AE and strain monitoring in laboratory-scale hydraulic fracturing experiments provides a promising approach to reduce uncertainties in fracture parameter estimation, thereby improving fracture network characterization and enabling real-time optimization of fracturing strategies.

In the study, we achieved joint monitoring of AEs and fracture attributes using different frequency bands of DAS data, and link the 3D distribution of AEs and relative strain as a basis for determining the orientation of fracture initiation. As a new geophysical monitoring tool, the wide-band response of DAS will be helpful for the identification and monitoring of fractures and AE activities.

Introduction

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In the study, we achieved joint monitoring of AEs and fracture attributes using different frequency bands of DAS data, and link the 3D distribution of AEs and relative strain as a basis for determining the orientation of fracture initiation. As a new geophysical monitoring tool, the wide-band response of DAS will be helpful for the identification and monitoring of fractures and AE activities.

Experimental setup

A rectangular semi-airtight stress-loading device ($2\text{ m} \times 2\text{ m} \times 1\text{ m}$) was used to couple artificial glutenite (comprising 66.5% gravel, with a Young's modulus of 15.36 GPa and a Poisson's ratio of 0.21) to the optical fiber (Figure 1). Figure 1a illustrates the placement of the optical fiber within the container using brackets to form the monitoring system. The external optical fiber entered the container from the same point and was wrapped around the three fracture wells (Figure 1b). of cement, granular aggregates and water) was steadily poured into the container. Subsequently, concrete (composed of cement, granular aggregates, and water) was steadily poured into the container. The rock sample was then left to rest for two weeks to allow full solidification. Figures 1c-d show the three-dimensional and top views of the actual DAS array after 20,000 interpolations, dividing the fiber into four external planes (Faces A–D) and one internal plane (Face I). Notably, the optical fiber cannot be bent at 90° angles or completely at the corners.

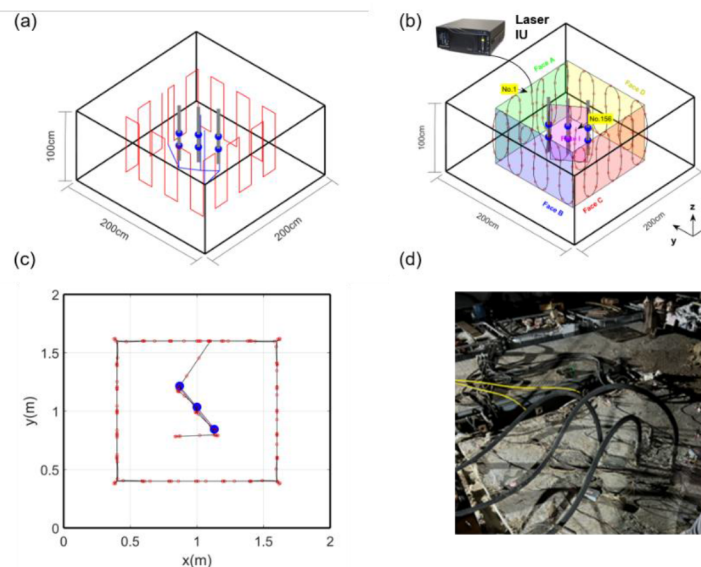


Figure 1—Geometry of concrete-coupled fiber-optic DAS monitoring system. (a) Theoretical fiber-optic deployment. (b) Actual fiber-optic deployment. (c) Site construction diagram shows three clear injection ports. (d) Top view of the actual DAS system.

Acoustic emission (AE) signals were recorded using a high-resolution fiber-optic Distributed Acoustic Sensing (DAS) system capable of detecting axial strain changes with sub-nanometer precision. The system provides broadband sensitivity, effectively capturing signals from below 0.001 Hz to 50 kHz and encompassing dynamic processes from slow fluid migration to rapid brittle failure. A 200-meter-long optical fiber was deployed in the experiment, configured with a 0.2 m channel spacing and a 2 m gauge length. A total of 156 DAS channels were recorded at a sampling rate of 40 kHz, distributed across five observation planes (denoted as Faces A–D and I). The fiber installation began at Face A and extended to Face I, as illustrated in Figure 1c. Due to the large size of the experimental rock sample, conventional ultrasonic monitoring was not feasible. However, pre-test wave velocity measurements were conducted along the specimen's central axis. The measured P-wave velocities ranged from 3200 m/s vertically to 2800 m/s

horizontally, indicating negligible seismic anisotropy. The specimen was visually intact, with no apparent surface fractures and a uniformly filled interior. For the purpose of AE event localization, a representative P-wave velocity of 3000 m/s was adopted.

Methodology

AEs localization based on high-frequency DAS data

The experiment produced 780 seconds of continuous acoustic emission (AE) waveform recordings. Discrete AE events were subsequently extracted using automated detection algorithms. The background noise level of each DAS channel was quantified by calculating the average strain-rate amplitude prior to event detection, which remained below $0.02 \text{ } \epsilon/\text{s}$. This value was adopted as the amplitude-based triggering threshold. A microseismic AE event was defined as occurring when at least 20 independent channels simultaneously exceeded the threshold within a 12 ms time window. For each identified event, a 51.2 ms waveform segment (corresponding to 2048 samples), centered on the trigger time, was extracted and archived. Based on this criterion, a total of 1,667 AE events were identified during the experiment.

The high spatial sampling density of DAS ensures strong waveform similarity among adjacent channels (Fig. 2a-c). To accurately detect P-wave arrivals of AE events, we propose a hybrid method combining the variance-based Akaike Information Criterion and cross-correlation (AIC-CC). The initial P-wave arrival on the first channel is determined by locating the minimum of the AIC curve (blue band in Fig. 2a). Cross-correlation is then applied to calculate the relative time shift between the first and the i -th channels, allowing estimation of absolute arrival times (blue bands in Fig. 2b and 2c). Figures 2d and 2e illustrate the P-wave arrival times and corresponding signal-to-noise ratios (SNR) across all 156 DAS channels. The 20 channels with the highest SNR (highlighted in red in Fig. 2e) were selected for subsequent localization analysis.

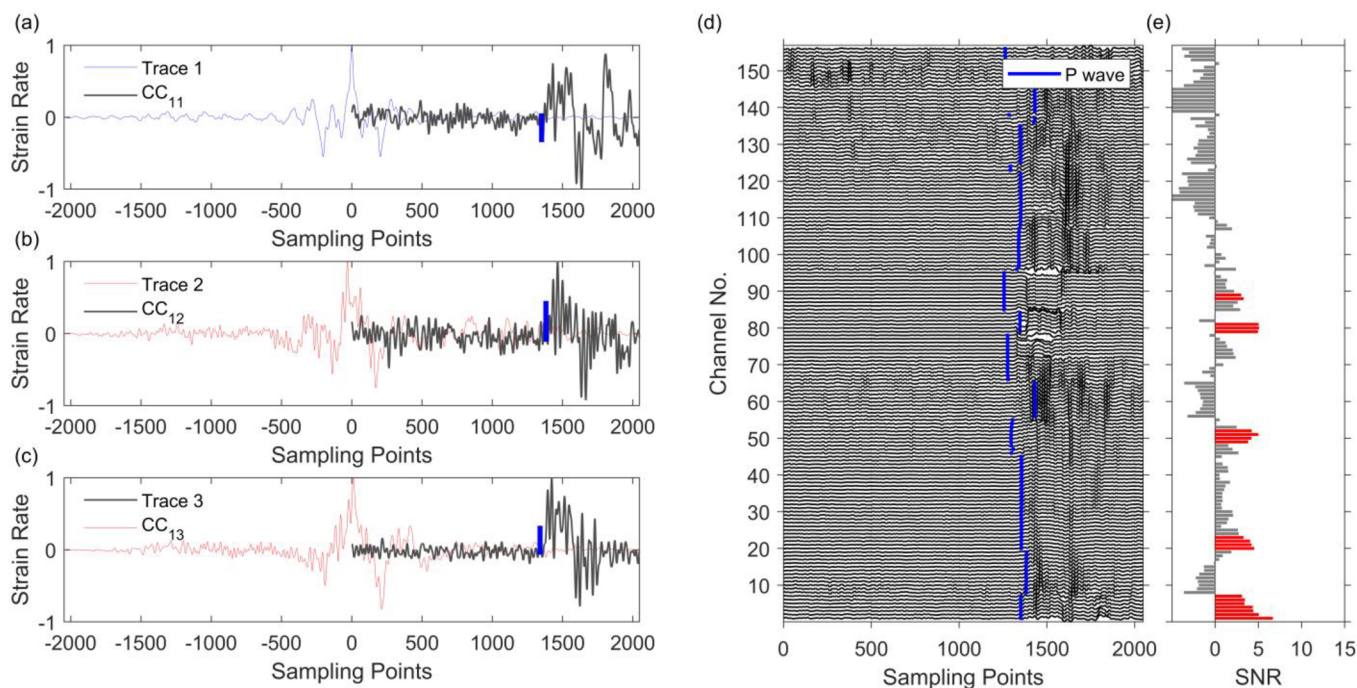


Figure 2—Example of P-wave arrival picking using the AIC-CC method. (a–c) AE waveforms (grey) and corresponding correlation curves (blue/red) from Channels 1, 10, and 100, respectively. (d) DAS record showing picked P-wave arrivals (blue bars) across all channels. (e) Signal-to-noise ratio (SNR) of each channel, with highest-SNR channels highlighted in red.

After picking P-wave first arrivals automatically, we then implement the collapsing grid search algorithm to locate these triggered events:

$$E_{RMS_i} = V_p \sqrt{\frac{\sum_{j=1}^N (\Delta T_{ij})^2}{N}} \quad (1)$$

where V_p is the P-wave velocity and N is the number of DAS channels, here $N=20$. ΔT_{ij} is the arrival time difference between the observed and the theoretical arrival computed by scanning the grid points of three-dimensional space domain. An optimal source with a minimum localized error (Root-Mean-Square error, E_{RMS}) is then computed from the residuals.

Fracture diagnosis based on Low frequency DAS data

Similar to the processing of high-frequency data, the raw DAS recordings were low-pass filtered below 1 Hz and downsampled to a resolution of 1 ms. A median filter was then applied to suppress high-amplitude noise. Given the direct connection between the laser interrogator and the DAS array, it was necessary to remove the DC offset introduced during demodulation. Figures 3a and 3b display the raw and low-frequency DAS (LF-DAS) data along the entire fiber length (200 m) during the interval from 480 to 600 seconds, with red and blue indicating tensile and compressive deformation, respectively. Figures 3a-1 and 3b-1 provide zoomed-in views of the boxed regions. Figure 3c illustrates the variation of injection pressure over time. The LF-DAS data clearly exhibit alternating tensile and compressive strain responses during pressure loading.

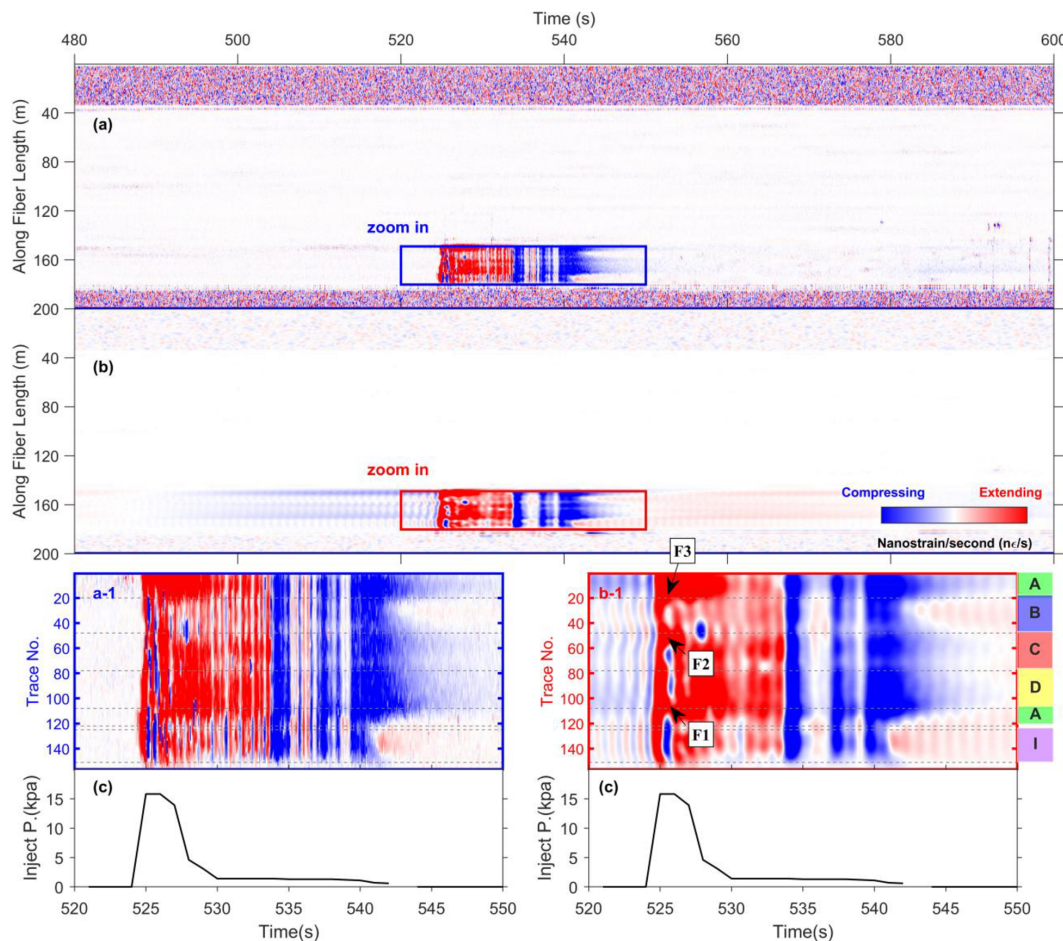


Figure 3—The Fiber-optic DAS monitoring of strain rate and pressure over time for 120 s before and after injection pressure loading (a) Raw DAS data, a-1 indicates a zoomed-in plot in the blue box. (b) LF-DAS, b-1 indicates a zoomed-in plot of the data in the red. (c) Injection pressure curves. We use Compressing and Extending to describe the strain rate in the figure, respectively, and a continuous tensile response and stress shadows can be seen in Figure 4b-1, which may correspond to three potential fractures.

When a fracture tip propagates toward the optical fiber, the DAS system generates a distinct strain-rate response, allowing the fracture direction to be qualitatively inferred from the signal characteristics of the LF-DAS data. If the fracture tip makes direct contact with the fiber, a pronounced tensile response appears in the contacted channels, accompanied by symmetric compressive responses on both sides—commonly referred to as heart-shaped patterns or stress shadows. In contrast, when no direct fracture–fiber interaction occurs, a similar symmetrical pattern appears across the nearest channels and their neighbors, forming what we term "butterfly diagrams."

In this study, a round-trip folding configuration was employed to increase the contact area between the fiber and the cement matrix. However, this design also enhanced spatial averaging effects, which weakened the resolution of lateral stress shadows. As a result, tensile responses were primarily detected only when the fractures were in close proximity to or directly intersected the fiber. Despite these limitations, three distinct tensile responses—presumably representing individual fractures—were clearly identified (Fig. 3b-1). Notably, at least three polarity reversals occurred during the period of decreasing injection pressure, suggesting rapid and transient opening–closure cycles in response to the combined effects of water injection and pressure drop.

Result

AE response: Interpretation of high-frequency DAS data

High-frequency DAS recordings of acoustic emissions (AEs) typically exhibit sharp, high-amplitude signals that stand out clearly from background noise, making them well-suited for resolving the spatiotemporal evolution of AE swarms. In this study, a collapsing grid search algorithm was employed to localize AE events within a theoretical volume of $2.0\text{ m} \times 2.0\text{ m} \times 1.0\text{ m}$. The resulting three-dimensional distribution of AE sources is shown in Figure 4a. Figures 4b-f and 4c display the cumulative AE event count and the corresponding histories of injection pressure and rate, respectively, revealing a distinct three-stage rupture process.

Stage 1 marks the onset of fracturing and is characterized by a relatively low AE rate of approximately 5 events per second. As shown in Figure 4c, this stage coincides with the constant-loading phase of confining pressure. Stage 2 represents the main rupture phase, during which AE activity intensifies significantly, ranging from 10 to 50 events per second, and includes at least two pronounced bursts of elevated emissions (Figure 4b). Injection pressure begins to rise at 524 s, peaks sharply at 15.8 kPa within 1 s, and then gradually declines to 1.4 kPa by 529 s, corresponding to the first rupture burst. By 541 s, the pressure further drops to 0 kPa, indicating the second rupture episode.

Notably, fluid injection did not cease immediately following the pressure drop; water continued to flow and passively migrated into the fracture system. During this phase, fluid movement was not accompanied by significant AE activity. Once the fractures became fully saturated, pore pressure began to rise again, initiating oscillatory tensile–compressive strain cycles that triggered renewed AE activity. Stage 3 is thus identified as the volumetric rupture phase, characterized by AE rates ranging from 5 to 30 events per second—likely associated with pore pressure diffusion throughout the fractured medium.

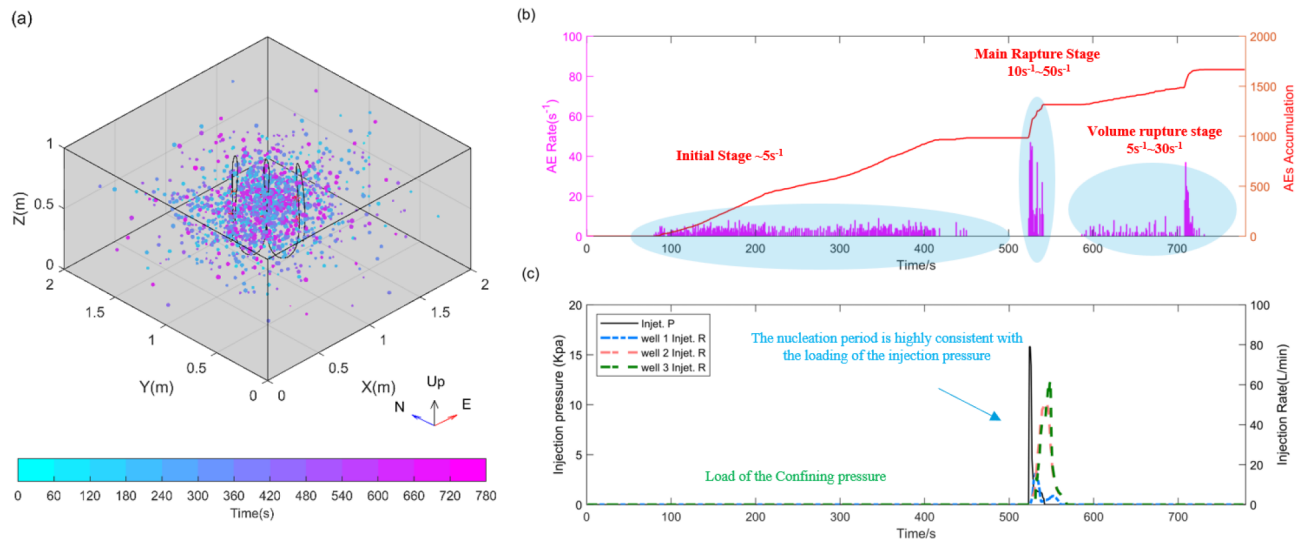


Figure 4—Spatiotemporal distribution of AE events and injection parameters. (a) 3D locations of AE events, with color indicating time and marker size scaled by relative magnitude. Axes point East (x), North (y), and Up (z). (b) AE rate and cumulative count over time, with blue markers indicating three rupture stages. (c) Injection pressure and rate over time at the three perforation points.

Figure 5 presents both three-dimensional and sectional views of AE event distributions during the three rupture stages. Stage 1 accounts for approximately 59% of the total AE events, with hypocenters clustered around the fractured wells. Over time, the event swarm exhibits spatial contraction, indicative of localized fracturing near the perforation zones. Stage 2 consists of 333 events ($\sim 20\%$ of the total), characterized by the outward migration of higher-magnitude events, primarily concentrated at depths between 0.4 and 0.6 m. Stage 3 comprises 350 events (21%), predominantly distributed around the fractured wells but with further outward expansion. This stage also exhibits a notable increase in high-magnitude events, which is likely attributable to fluid-induced seismicity.

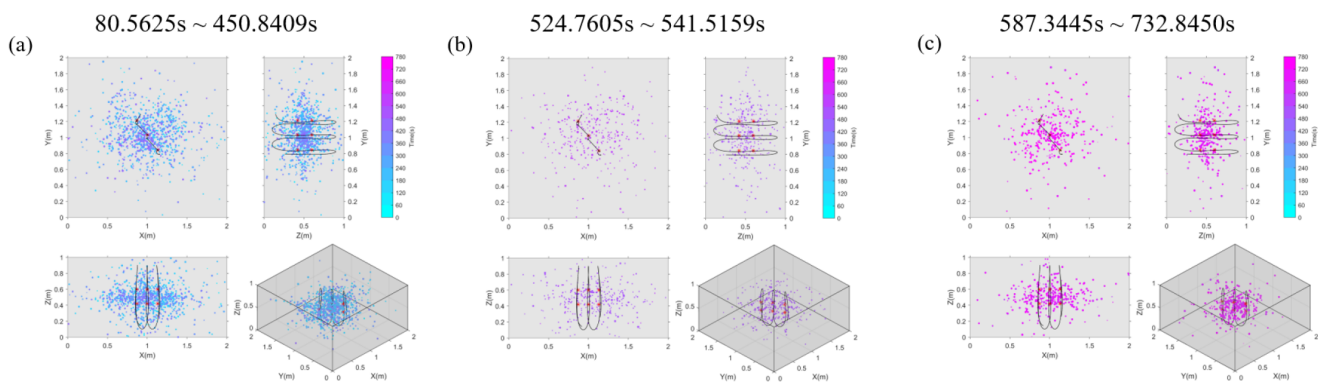


Figure 5—2D slices and 3D distributions of located AE events for each rupture stage: (a) initial stage, (b) main rupture stage, and (c) volumetric rupture stage. Stage durations are indicated above each subplot. Event markers are color-coded by time and scaled in size by relative magnitude.

These observations suggest that numerous microfractures were likely initiated around the six perforation points (Figure 1c) during the main rupture stage. The spatial distribution of AE events across the three stages exhibits strong similarity, indicating a persistently active fracturing zone. However, AE events occurring after the main rupture tend to exhibit higher magnitudes and reduced spatial clustering.

Overall, AE activity remains concentrated around the perforated sections of the fractured wells. To further characterize the geometry and evolution of the fracture network, low-frequency DAS data are required to complement the high-frequency AE observations.

Fracture response: Interpretation of low-frequency DAS data

This study aims to characterize the dynamic behavior of fracture opening and closure, as well as to qualitatively infer fracture orientations, using low-frequency distributed acoustic sensing (LF-DAS) signals. To this end, the optical fiber was divided into two groups of observation planes: the outer faces (A, B, C, and D) and the inner face (I). The LF-DAS responses from these two groups were analyzed independently.

Figure 6 illustrates the three-dimensional LF-DAS response along the outer fiber faces during the initial 2 seconds following water injection (524–526 s), recorded at a temporal resolution of 0.2 s. A distinct tensile signal first appeared along the northeastern segment of the fiber (Figure 6a) and then sequentially propagated to adjacent outer faces (Figures 6b–6f). As deformation progressed, directional attenuation and polarity reversals were observed in several channels. This transition is characterized by a weakening of tensile signatures along the western and southeastern segments, accompanied by the emergence of compressional responses in previously tensile-dominated northeastern channels (Figures 6g–6h), suggesting a temporal shift in fracture behavior and propagation direction.

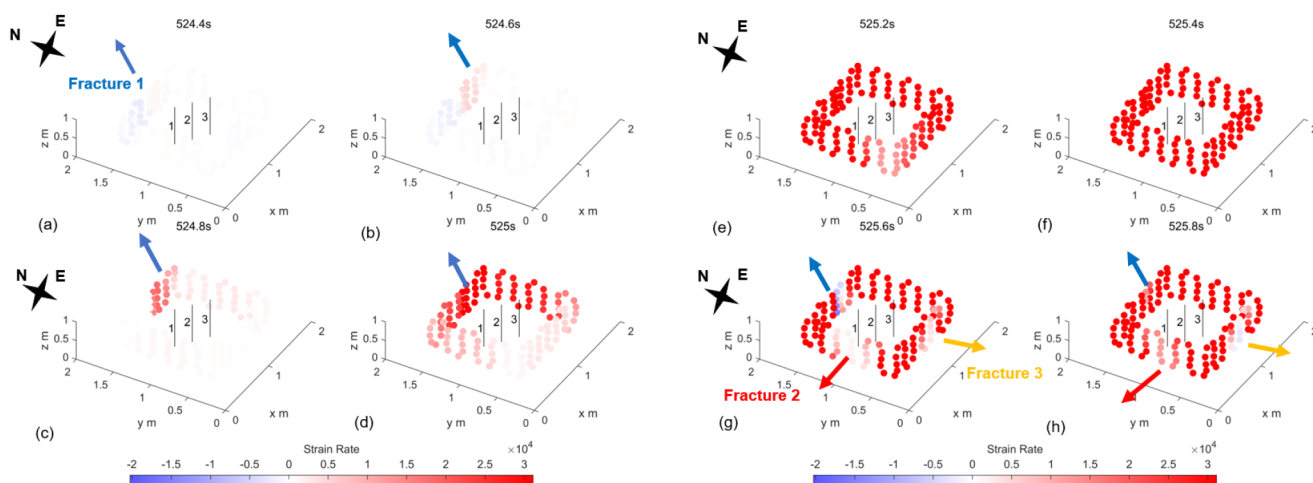


Figure 6—3D LF-DAS responses on the outer fiber faces (Faces A–D) within 2 seconds after water injection. Snapshots are shown at (a) 524.4 s, (b) 524.6 s, (c) 524.8 s, (d) 525.0 s, (e) 525.2 s, (f) 525.4 s, (g) 525.6 s, and (h) 525.8 s. The blue, red, and yellow arrows indicate the directions of three dominant fractures. Red and blue circles represent tensile and compressive responses, respectively, for each channel. The positive x- and y-axes point East and North, respectively.

Figure 7 illustrates the three-dimensional LF-DAS response on the inner fiber face (Face I) during the first 2 seconds following water injection (524–526 s). As shown in Figures 7a–d, a tensile signal initially appeared near Shot 1 and gradually intensified toward Shot 3. When combined with the simultaneous LF-DAS response observed on the outer fiber (Figures 6a–d), this pattern indicates the development of a northeast-trending fracture propagating outward from the perforation zone.

Subsequently, the tensile signal continued to strengthen toward Shot 3 (Figures 7e–f), suggesting the possible initiation of an east–west oriented fracture. However, after reaching its peak, the signal rapidly weakened and reversed polarity (Figure 7g), indicating a sudden closure of this fracture. By the end of the observation period (Figures 7g–h), tensile responses near Shots 1 and 2 had transitioned into compressional signals, while only Shot 3 retained a weak tensile amplitude. This transition coincided with an AE burst peaking at 50 events per second, suggesting the formation of a northwest–southeast oriented fracture that may have intersected the perforation cluster and contributed to the observed polarity reversals. It is important to note that this experiment simulated water injection without the use of proppants, which may have facilitated multiple rapid cycles of fracture opening and closure.

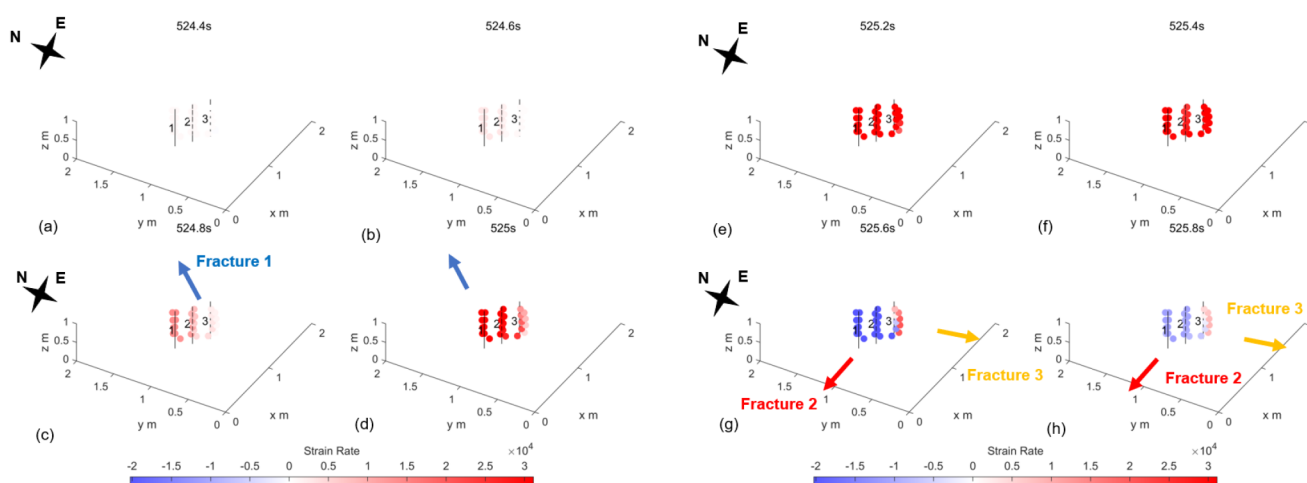


Fig 7—3D LF-DAS responses on the inner fiber face (Face I) within 2 seconds after water injection. Snapshots are shown at (a) 524.4 s, (b) 524.6 s, (c) 524.8 s, (d) 525.0 s, (e) 525.2 s, (f) 525.4 s, (g) 525.6 s, and (h) 525.8 s. Symbols are consistent with those in Figure 8, where red and blue circles denote tensile and compressive responses, respectively, and arrows indicate the inferred fracture orientations.

Based on the LF-DAS data (Figure 3b) and the spatial responses observed on both the outer (Figure 6) and inner (Figure 7) fiber faces, three dominant fractures can be qualitatively identified: (1) a southwest–northeast trending fracture (Fracture 1, indicated by blue arrows in Figures 6 and 7); (2) an east–west trending fracture (Fracture 2, red arrows); and (3) a northwest–southeast trending fracture (Fracture 3, orange arrows). Fractures 1 and 2 exhibit both tensile and compressive phases, enabling their propagation paths to be tracked through polarity reversals. In contrast, Fracture 3 appears to rupture in close proximity to the perforation zone, resulting in limited directional resolution. The observed sequence of polarity changes suggests that fracture closure during rupture may promote hydraulic connectivity among multiple fracture planes, ultimately leading to the formation of a complex fracture network around the perforations.

Conclusion

This study demonstrates the integrated application of Distributed Acoustic Sensing (DAS) across multiple frequency bands for laboratory-scale hydraulic fracturing monitoring. Using a cement-bonded fiber-optic DAS array, we recorded 1,667 effective acoustic emission (AE) events, which were categorized into three distinct stages corresponding to confining pressure loading, injection pressure increase, and fluid diffusion.

The cumulative AE event distribution follows a power-law relationship, with a measured b -value of 1.20—consistent with values typically observed in seismicity induced by hydraulic fracturing. Low-frequency DAS data enabled the qualitative identification of fracture orientations, revealing three dominant fracture sets: one striking southwest–northeast, another east–west, and the third northwest–southeast. Among these, the third set appears to serve as the principal fracture, potentially interacting with secondary fractures to form a complex fracture network near the perforation zones.

These findings underscore the potential of DAS as a comprehensive tool for high-resolution, real-time monitoring of fracture processes during hydraulic fracturing. Future research should aim to enhance spatial resolution, integrate multi-physics datasets, and scale the methodology to field applications in order to improve fracture diagnostics and optimize stimulation strategies.